Variational Quantum Bottleneck Conditioning for Speech Enhancement under Noise and Speaker Shift

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Abstract

One of the ways to incorporate quantum feature maps into a speech-enhancement network is through global conditioning. We proposed a four-layer, six-qubit variational quantum circuit that transforms multiscale acoustic features into twelve-qubit and pairwise moments, which later modulate a phase-aware U-Net bottleneck. The classical conditioner shares the same architecture as well as the same parameter count: the complete network contains 529,823 parameters, and the quantum and classical core contains 72 trainable scalars. Bottleneck modulation helps us to characterize the encoded angle Fourier support, propagation of moment estimation error, and shared measurement statistics. Experimenting on two 267-mixture evaluations, seed values of 17, 23, and 123 report positive mean quantum minus classical difference in SI-SDR, PESQ, and ESTOI. Unseen-noise gains average 0.04972 dB, 0.01240 and 0.000885, respectively; jointly unseen speakers and noise yield gains of 0.05703 dB, 0.01256 and 0.000588. The findings show a quantum conditioning mechanism and consistent directions of average improvement within the reported runs, motivating broader evaluation of structured quantum feature maps for speech enhancement.

Keywords: variational quantum circuits; speech enhancement; data re-uploading; quantum measurements; bottleneck conditioning; distribution shift.

1 Introduction

The single-channel speech enhancement task is defined as the recovery of a clean speech signal from a recording mixed with the acoustic interface. This task is divided into two subtasks: one is to suppress unwanted sound or noise, and the second is to retain the spectral and temporal structure that leads to natural and intelligible speech. Balance between these two subtasks is relevant when training speech data differs from the testing speech data. The model needs to be generalized so that it is applicable to unfamiliar interfaces as well as unfamiliar speakers. With this setting, the question is not only how accurately a network learns the spectrum, but also how its features can adapt to the acoustic context of input speech.

This kind of problem can be solved using complex-spectrum and phase-aware models. Using different architectural choices of Deep Complex U-Net, PHASEN, and DCCRN, phase-sensitive information can be enhanced [1, 2, 3]. This motivates us to extract the kind of features that retain both real and imaginary components or phase coordinates and related amplitudes. In the case of an encoder-decoder network, local features explain the time-frequency structure, while the encoder representation combines information over larger regions. This architecture helps to connect detailed reconstruction with a compact summary of the segment being processed. This study focuses on that connection using a learnable conditioning module at the bottleneck of the U-Net.

The definition of bottleneck conditioning in this case is to adjust the network's feature channels using the input data itself. The bottleneck works as a compressed feature extraction between the encoder and decoder. This helps to supply input-dependent scale and bias values that modify these features before reconstruction. These values change between input segments and are shared with each channel across the time-frequency positions. This opens access to a global acoustic descriptor while keeping the spatial feature maps, and skip connections are used to reconstruct speech. The main idea is related to channel recalibration and feature-wise affine conditioning in classical neural networks [16, 17].

Our descriptor is a combination of channel-wise means, four channel standard deviations, and the bottleneck, which gives us complementary information about average feature activation and its variation within the segment. The descriptor can incorporate features learned at different levels of the hierarchy with the help of pooling multiple scales. 896-dimensional feature vector is then projected to six bounded

coordinates, which is central to the proposed hybrid architecture: acoustic structure is extracted by a classical encoder, and nonlinear mapping provides coefficients that control bottleneck modulation.

A variational quantum circuit is one way to do nonlinear mapping. Input values can be encoded as a rotational angle for a quantum circuit, which is later transformed using unitary gates and read out through an expectation value. During circuit evolution, data re-uploading introduces the same input [**?**]. To understand the representational role of encoded coordinates, Fourier dependence of quantum expectation values is used, where repeated encoding can enlarge the accessible frequency support [6]. This leads to using a quantum circuit as a feature transformation whose architecture is defined by the rotational quantum gates, encoding, and measurement operator. Our goal is to study how this transformation behaves for the speech enhancement network.

Our proposed conditioner quantum circuit architecture has six qubits and four circuit layers. Learned coordinates are re-encoded at each layer, along with trainable single-qubit rotational gates, and qubits are connected through a CNOT gate ring. To compute the expectation value, a single-qubit Z gate is used, and six adjacent ZZ moments are computed, which result in twelve bounded feature values. This computed expectation value represents the circuit's computational basis probability distribution. Later, these expectation values are converted into channel-wise scale and bias values, and to control the strength of the bottleneck of these values, bounded modulation is used. The classical backbone performs the time-frequency reconstruction, while the quantum variational circuit acts on a global summary for every processed segment.

This architecture makes the relationship between acoustic processing and quantum processing obvious. Our interface uses pooled statistics, due to which the number of time-frequency bins changes, whereas the number of circuit input coordinates is fixed. To compute the expectation value of the quantum circuit, a differentiable statevector simulator is used, while the complete segment still passes through the classical network. This results in an architectural benefit, which is a close-packed and organized conditioning mechanism. To evaluate empirical values, we rely on the enhanced speech generated by the quantum-classical hybrid model, while its mathematical properties are evaluated through the precise quantum variational circuit and its expectation values.

A quantum vs. classical comparison is a must to interpret this analysis. To compare with quantum, we construct a classical nonlinear core having six inputs and twelve outputs, which uses a two-layer bias-free map with a fixed pair product growth. Both classical and quantum store 72 trainable scalars, and both networks contain 529,823 parameters, along with the same bottleneck modulation rules. This comparison is interpreted in terms of stored parameter count and interface. In the methodology, we have specified effective parameter dependence and training settings differently so that the experimental configuration is reproducible.

Parameterized quantum circuits have a direct connection between state representation and measurement. While considering the computational basis, all twelve readout observables are diagonal in the computational basis; this allows them to be evaluated from the same expectation values in a prospective sampled implementation. Due to their shared origin, we can determine both the correlation between estimates and individual estimation variance. Using a bounded modulation rule, we observed these quantities and propagated feature-estimation error. For each repeated encoded input angle, we applied gate-wise parameter-shift differentiation. These results help us to understand what the conditioner computes, how expectation values affect the classical feature transformation, and how its derivatives are formed.

For the experimental study, we have considered: unseen noise with speakers represented during training, and a joint change in speaker identity and noise. For evaluation, SI-SDR, PESQ, and ESTOI are used to describe waveform fidelity, perceptual quality, and intelligibility-related behavior through complementary objective measures. For each reported seed—17, 23 and 123—the quantum-conditioned network achieves higher mean scores than the classical-conditioned network on all three measures in both evaluations. We report absolute scores, paired differences, and variation across the reported runs.

Our contributions are

- We proposed a quantum conditioner that connects bounded modulation of a phase-aware U-Net to multiscale acoustic statistics, along with a classical core having the same number of stored parameters.
- We proposed quantum-model principles to derive properties of the circuit's frequency support, expectation value statistics, differentiation, and readout error.
- We document positive average enhancement differences in the reported runs under two distribution shifts and provide independent numerical checks of the quantum implementation.

2 Related work and positioning

2.1 Phase-aware speech enhancement

Phase-aware enhancement provides the acoustic foundation for the present model. Deep Complex U-Net develops complex-valued processing for speech enhancement [1]. PHASEN uses communicating amplitude and phase streams and frequency transformation blocks to capture harmonic relationships [2]. DCCRN combines complex convolutional processing with recurrent modeling to handle complex spectral targets [3]. These approaches demonstrate different ways of representing and reconstructing speech when phase information is included. Our backbone follows this broader motivation by retaining phase-related input coordinates and predicting a complex enhanced spectrum. Its role is to provide a common reconstruction system within which the conditioning map can be studied.

For evaluation, the MetricGAN+ criterion is used, which develops a framework for learning with perceptually motivated metric feedback [4], illustrating the relevance of quality-oriented objectives alongside signal reconstruction. In our experiments, SI-SDR, PESQ, and ESTOI give complementary views of the output [11, 14, 15]. Experiments contain a predefined synthetic-mixture protocol, and results are analyzed within that protocol. For related context, results from VoiceBank+DEMAND are used, while direct numerical comparisons need to match data construction, evaluation setting, and partitioning.

2.2 Global feature conditioning and channel modulation

To establish a useful architectural reference, classical conditioning methods are used. Through conditioning information, FiLM generates feature-wise transformations, giving control of one presentation over another [16]. To recalibrate feature responses, the classical squeeze-and-Excitation network uses learned channel dependencies and global aggregation [17]. Both of these approaches treat the global descriptor as a source of channel-level adaptation. In our architecture, the descriptor is derived internally from multiscale acoustic features, and the resulting scale and bias modulate the U-Net bottleneck.

The question we focused on deals with the nonlinear transformation between the modulation and the global descriptor. With the same scalar counts, we use a learned six-coordinate projection followed by either a variational quantum circuit or a classical core. The same bounded modulation is fed to both. This construction connects the study to established conditioning methods while choosing a core explicit.

2.3 Quantum models for acoustic processing

QSpeech works on a quantum module for speech applications with few qubits [8], which establishes an early reference for using quantum processing in acoustic tasks. HQCSE is a hybrid, quantum-plus-classical approach for the application of speech enhancement and was experimented on TIMIT-derived and VoiceBank+DEMAND data [9]. QASA-SincNet is a four-qubit attention module within a complex-mask enhancement framework [10]. The results of these studies establish that quantum-classical speech processing is an active area of research and motivate us to find which architectural roles are appropriate for a small circuit.

We focus on multiscale global statistics, which contain a single segment-level bottleneck quantum variational circuit and a shared classical comparison. It helps in the specified integration of these components and the investigation of the expectation value and modulation. The quantum circuit is a six-qubit variational circuit, which gives the expectation value upon measurement, while the encoder and decoder control the responsibility for acoustic feature extraction and reconstruction. This gives us an opportunity to examine the output as measurable conditioning features.

2.4 Representational structure, differentiation and measurement

Data re-uploading alternates repeated input encoding with trainable circuit operations [5]. The Fourier analysis of quantum models relates accessible frequencies to the spectra of encoding generators and to how often encoding occurs [6]. We apply this framework to four repetitions of six Pauli-generated input rotations and obtain an upper bound on frequency support in the encoded coordinates. This characterization specifies a structural property of the chosen circuit; the actual coefficients depend on its trainable gates and readout.

Applying parameter-shift supplies expectation-value derivatives for separated Pauli rotation gates [7]. Due to re-uploading, each input occurs in several gates, so the resulting derivative comprises contributions

from all occurrences. Our method checks the derivatives numerically. We also investigate the joint probabilities of the commuting Z and ZZ expectation values and the impact of their estimation error on bottleneck modulation. We connect quantum principles to the precise engancement architecture using application-specific derivations.

3 Problem formulation and enhancement backbone

3.1 Signal representation

Let $s[t]$ denote the clean speech and $n[t]$ denote a noise recording. In the case of nonzero noise power P_n and target SNR γ in decibels, define

$$x[t] = s[t] + a_n n[t], \qquad a_n = \sqrt{\frac{P_s}{P_n 10^{\gamma/10}}}. \tag{1}$$

Joint rescaling of the resulting mixture and clean target preserves the target SNR. Let X_{fk} and S_{fk} be their short-time Fourier transforms. The analysis configuration uses a 320-point FFT, a 320-sample window, and a 160-sample hop at 16 kHz. The enhancement target is a complex spectrum $\widehat{S}$ and its inverse-STFT waveform $\widehat{s}$.

We denote the stabilized magnitude $m_{fk} = \sqrt{(\Re X_{fk})^2 + (\Im X_{fk})^2 + \epsilon}$, with $\epsilon = 10^{-8}$, and $a = \max(\max_{f,k} m_{fk}, \epsilon)$. The scale is separated from automatic differentiation. The five-channel representation is

$$\mathbf{F}_{fk} = \left[\frac{\Re X_{fk}}{a}, \frac{\Im X_{fk}}{a}, \frac{\log(1+m_{fk})}{\max(\log(1+a), \epsilon)}, \frac{\Im X_{fk}}{\max(m_{fk}, \epsilon)}, \frac{\Re X_{fk}}{\max(m_{fk}, \epsilon)}\right]^\top. \tag{2}$$

Here, the last two coordinates define phase through a bounded Cartesian ratio without requiring any phase unwrapping. Consider these as stabilized phase coordinates, instead of exact sine and cosine at bins with negligible magnitude. The STFT uses a periodic Hann window, centered frames, and reflection padding; the inverse transform restores the requested waveform length.

3.2 Shared backbone and global summary

The encoder channel widths are 32, 64, 96, and 128. The 128-channel bottleneck contains three residual depthwise-separable gated blocks with dilations 1, 2, and 4. The decoder restores resolution, combines skip features, and predicts four reconstruction maps. Dropout is 0.08. This backbone and the conditioner interface are shared by the classical and quantum arms.

In case of a tensor feature $H \in \mathbb{R}^{C\times T\times F}$ is defined by channel-wise statistics

$$\mu_c(H) = \frac{1}{TF}\sum_{t,f} H_{ctf}, \qquad \sigma_c(H) = \sqrt{\frac{1}{TF}\sum_{t,f}(H_{ctf} - \mu_c(H))^2}. \tag{3}$$

The population convention is used for the standard deviation without any added stabilizer. Implementation default $\epsilon_{\mathrm{LN}} = 10^{-5}$ is used for layer normalization. Concatenating means and standard deviations from the four encoder stages and the bottleneck gives $\mathbf{g} \in \mathbb{R}^{896}$, since $2(32+64+96+128+128) = 896$. The conditioner input is

$$\boldsymbol{\varphi} = \pi \tanh(\mathrm{LN}(W_{\mathrm{in}}\mathbf{g} + \mathbf{b}_{\mathrm{in}})) \in [-\pi, \pi]^6. \tag{4}$$

The quantum operation receives six learned global coordinates summarizing the segment. Its input dimension is independent of the segment's time–frequency grid size.

4 Quantum and classical conditioning maps

4.1 Ordered variational quantum circuit

Considering $q = 6$ and $L = 4$, we define the Pauli rotations $R_A(\beta) = \exp(-i\beta A/2)$ and the encoding block

$$D(\boldsymbol{\varphi}) = \bigotimes_{i=1}^{q} R_Y(\varphi_i). \tag{5}$$

At layer ℓ, the local trainable block is

$$V_\ell(\boldsymbol{\theta}_\ell) = \bigotimes_{i=1}^{q} R_Z(\theta_{\ell i3}) R_Y(\theta_{\ell i2}) R_Z(\theta_{\ell i1}). \tag{6}$$

Rightmost operators act first. Let C_i be the CNOT from i to $i+1$ for $i < q$, and C_q the CNOT from q to 1. The ordered ring is $E = C_q C_{q-1} \cdots C_1$. The full state is

$$|\psi(\boldsymbol{\varphi}, \boldsymbol{\theta})\rangle = U_L \cdots U_1 |0\rangle^{\otimes q}, \qquad U_\ell = E V_\ell D(\boldsymbol{\varphi}). \tag{7}$$

The chronological local gate sequence is $R_Y(\varphi_i)$, $R_Z(\theta_{\ell i1})$, $R_Y(\theta_{\ell i2})$, $R_Z(\theta_{\ell i3})$, followed by the ordered CNOT ring. The evaluated architecture uses one circuit at the bottleneck.

The feature vector contains six single-qubit means and six cyclic pair moments,

$$\mathbf{z}_Q = \big(\langle Z_1\rangle, \ldots, \langle Z_q\rangle, \langle Z_1 Z_2\rangle, \ldots, \langle Z_q Z_1\rangle\big)^\top \in [-1, 1]^{12}. \tag{8}$$

All expectations use the state in Eq. (7). The reference implementation computes these expectations by summing computational-basis probabilities in a differentiable complex64 statevector simulator. The empirical study concerns ideal, exact-expectation simulation subject to floating-point precision. Section 5 additionally characterizes a prospective shared-shot measurement implementation.

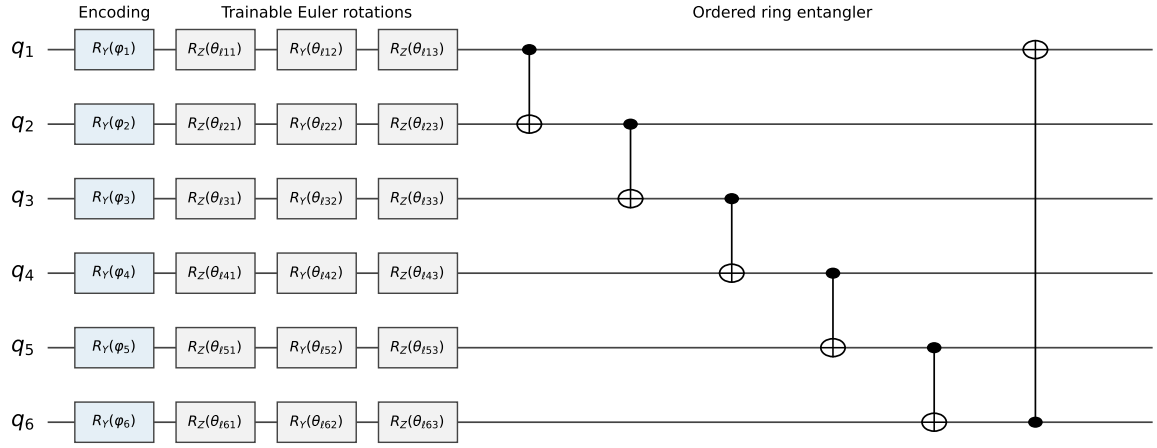


Figure 1: One layer of the defined circuit, in chronological order from left to right. The same six input angles are re-uploaded in each of four layers. Computational-basis measurement is performed after the final layer, not between layers. The wrap-around CNOT completes the ordered ring.

4.2 Classical core with the same stored scalar count

The classical comparison is a bias-free $6 \to 6 \to 6$ nonlinear map followed by a fixed expansion:

$$\mathbf{h} = \tanh\big(W_2 \tanh(W_1 \boldsymbol{\varphi})\big), \qquad W_1, W_2 \in \mathbb{R}^{6\times 6}, \tag{9}$$

$$\mathbf{z}_C = (h_1, \ldots, h_6, h_1 h_2, \ldots, h_6 h_1)^\top. \tag{10}$$

It stores $2 \cdot 6^2 = 72$ weights; the VQC stores $3qL = 72$ angles. The fixed product expansion supplies the twelve-dimensional interface without extra trainable weights. This comparison matches stored count and interface while allowing each core its own nonlinear representation.

4.3 Shared bottleneck modulation

For either $\mathbf{z} = \mathbf{z}_Q$ or $\mathbf{z}_C$, let

$$\mathbf{a} = \tanh(W_a \mathbf{z} + \mathbf{b}_a), \qquad \mathbf{b} = \tanh(W_b \mathbf{z} + \mathbf{b}_b), \qquad \alpha = \tfrac{1}{4}\,\mathrm{sigmoid}(\kappa), \tag{11}$$

where $W_a, W_b \in \mathbb{R}^{128\times 12}$. The conditioned bottleneck is

$$\widetilde{H}_{ctf} = H_{ctf}(1 + \alpha a_c) + \alpha b_c. \tag{12}$$

In particular, $0 < \alpha < 1/4$, $1 - \alpha < 1 + \alpha a_c < 1 + \alpha$ and $|\alpha b_c| < \alpha$. The affine modulation is bounded for a fixed bottleneck, and the quantum core is invoked once per processed segment at the model-interface level. Hardware execution would require repeated state preparations to estimate expectations.

The conditioner parameter count is

$$\underbrace{896 \cdot 6 + 6}_{\text{input projection}} + \underbrace{2 \cdot 6}_{\text{affine LN}} + \underbrace{72}_{\text{core}} + \underbrace{12 \cdot 256 + 256}_{\text{output projection}} + \underbrace{1}_{\text{strength}} = 8795. \tag{13}$$

The 521,028-parameter backbone and 8,795-parameter conditioner give 529,823 stored trainable scalars in either conditioned model. These totals include both projections and affine layer normalization and are confirmed by module-level counting.

5 Mathematical properties of the quantum conditioner

5.1 Fourier structure of repeated angle encoding

Proposition 1 (Finite input-frequency support)**.** *For Eq.* (7)*, with trainable angles fixed, each output coordinate has an expansion*

$$z_k(\boldsymbol{\varphi}) = \sum_{\boldsymbol{\omega}\in\{-L,\dots,L\}^q} c_{k,\boldsymbol{\omega}}(\boldsymbol{\theta}) e^{i\boldsymbol{\omega}^\top \boldsymbol{\varphi}}, \qquad c_{k,-\boldsymbol{\omega}} = c^*_{k,\boldsymbol{\omega}}. \tag{14}$$

This is a support upper bound; it does not imply that every frequency is present or independently controllable.

Proof. For the encoding generator $Y/2$, the eigenvalues are $\pm 1/2$. During ket evolutions, each input coordinate occurs L times. Spectrally expanding encoding gates provides frequencies that are addition of L such eigenvalues. Each coordinate frequency is the difference of two sums, because the computed expectation value pairs a ket amplitude with a conjugate bra amplitude. □

For the selected circuit, $|\omega_i| \le 4$. This specialization follows the encoding-frequency viewpoint of Ref. [6]. It applies in the encoded coordinates $\boldsymbol{\varphi}$; after composing with Eq. (4), it is not a finite-Fourier claim about the waveform itself. The coefficients are constrained by the circuit and should not be treated as freely chosen classical Fourier coefficients.

5.2 Joint moments and shared measurements

Proposition 2 (A single measurement basis suffices)**.** *Let $B^{(s)} \in \{0,1\}^q$ be N independent computational-basis samples from the circuit. Define $v_i(B) = (-1)^{B_i}$ and $v_{q+i}(B) = (-1)^{B_i + B_{i+1}}$, using cyclic indices. Then*

$$\widehat{z}_k = \frac{1}{N}\sum_{s=1}^{N} v_k(B^{(s)}), \quad \mathbb{E}\widehat{z}_k = z_k, \quad \mathrm{Var}(\widehat{z}_k) = \frac{1 - z_k^2}{N} \le \frac{1}{N}. \tag{15}$$

Moreover,

$$\mathrm{Cov}(\widehat{z}_j, \widehat{z}_k) = \frac{\mathbb{E}[v_j(B)v_k(B)] - z_j z_k}{N}, \tag{16}$$

and, for $\epsilon > 0$,

$$\Pr(\|\widehat{\mathbf{z}} - \mathbf{z}\|_\infty \ge \epsilon) \le 24 e^{-N\epsilon^2/2}. \tag{17}$$

Proof. While computing the expectation value, every observable is diagonal, and each v_k is its ± 1 eigenvalue on the sampled bitstring. While considering the covariance of the two sample means, terms of different shots cancel each other, so the N equal-shot terms give Eq. (16). For each coordinate, we get $2e^{-N\epsilon^2/2}$ due to Hoeffding's inequality for a variable in $[-1, 1]$. Without assuming independence between coordinates, a union bound over twelve coordinates gives Eq. (17). □

Thus $N \geq 2\epsilon^{-2}\log(24/\delta)$ suffices for simultaneous accuracy ϵ with probability at least $1-\delta$. The covariance in Eq. (16) specifies how the shared measurement record couples the moment estimates.

The measured pair moments may contain connected correlations :

$$K_i = \langle Z_i Z_{i+1}\rangle - \langle Z_i\rangle\langle Z_{i+1}\rangle. \tag{18}$$

The classical output construction enforces $K_i = 0$ when interpreted as independent binary moments with means h_i. The quantum construction does not impose that factorization. However, these measured correlations are not, by themselves, entanglement witnesses. Dephasing the output state gives the separable state

$$\rho_{\mathrm{diag}} = \sum_{b\in\{0,1\}^q} |\langle b|\psi\rangle|^2 |b\rangle\langle b|, \tag{19}$$

which reproduces every diagonal observable in Eq. (8). The relevant representational distinction is the learned input-to-moment map and its ability to supply non-factorized features to the classical projection.

5.3 Propagation of measurement error through conditioning

Proposition 3 (Conditional bottleneck perturbation bound)**.** *Hold H, W_a, W_b, their biases and α fixed. Let $\widetilde{H}$ and $\widehat{\widetilde{H}}$ be Eq.* (12) *evaluated at $\mathbf{z}$ and $\widehat{\mathbf{z}}$. Write*

$$M = \max_{c,t,f} |H_{ctf}|, \qquad A = \max_{c} \sum_{k} |(W_a)_{ck}|, \qquad B = \max_{c} \sum_{k} |(W_b)_{ck}|. \tag{20}$$

Then

$$\max_{c,t,f} |\widehat{\widetilde{H}}_{ctf} - \widetilde{H}_{ctf}| \leq \alpha(MA+B)\|\widehat{\mathbf{z}} - \mathbf{z}\|_\infty. \tag{21}$$

The same upper bound holds for $\|\widehat{\widetilde{H}} - \widetilde{H}\|_F/\sqrt{CTF}$.

Proof. Since $|\tanh u - \tanh v| \leq |u-v|$, the scale and bias errors satisfy $|\widehat{a}_c - a_c| \leq \|(W_a)_{c,:}\|_1 \|\widehat{\mathbf{z}} - \mathbf{z}\|_\infty$ and the analogous inequality for b_c. Subtracting the two modulation formulas, applying the triangle inequality and then maximizing over the indices gives Eq. (21). The normalized Frobenius norm cannot exceed the largest entry magnitude. □

Corollary 1 (A sufficient shot budget for bottleneck accuracy)**.** *For $\tau > 0$ and $0 < \delta < 1$, when $\alpha(MA+B) > 0$, a sufficient condition for both bottleneck error measures in Proposition 3 to be at most τ with probability at least $1-\delta$ is*

$$N \geq \frac{2\alpha^2(MA+B)^2}{\tau^2}\log\frac{24}{\delta}. \tag{22}$$

Proof. Substitute $\epsilon = \tau/[\alpha(MA+B)]$ in Proposition 2. If the factor is zero, Eq. (21) makes the bottleneck error identically zero. □

This architecture result gives a satisfactory measurement budget for a chosen internal activation accuracy, conditional on a fixed model and segment.

5.4 Derivatives and effective parameter accounting

For a stored angle occurring in one Pauli rotation, the feature derivative obeys [7]

$$\frac{\partial z_k}{\partial \theta_j} = \frac{z_k(\theta_j + \pi/2) - z_k(\theta_j - \pi/2)}{2}. \tag{23}$$

By contrast, an input φ_i occurs in every encoding layer. Introduce separate copies $\varphi_{\ell i}$ before tying them. The correct shared-input derivative is

$$\frac{\partial z_k}{\partial \varphi_i} = \sum_{\ell=1}^{L} \frac{\partial z_k}{\partial \varphi_{\ell i}}\bigg|_{\varphi_{1i}=\cdots=\varphi_{Li}=\varphi_i}. \tag{24}$$

Above, each term reveals a two-shift rule, but shifting all occurrences simultaneously does not follow Eq. (23). Classical chain rules then connect these feature derivatives to the conditioner projections and enhancement loss.

Proposition 4 (Stored count versus output-relevant angles)**.** *For the ordered ansatz in Eq.* (7)*, all outputs in Eq.* (8) *are independent of the six angles* θ_{Li3}*. Accordingly, the stored count is 72, but at most 66 angle coordinates can influence these outputs.*

Proof. Write the final state as $ED_Z|\chi\rangle$, where $D_Z = \bigotimes_i R_Z(\theta_{Li3})$. Every readout O is diagonal. The CNOT network E permutes computational-basis states, so $E^\dagger OE$ is diagonal and commutes with D_Z. Hence

$$\langle\chi|D_Z^\dagger E^\dagger OED_Z|\chi\rangle = \langle\chi|E^\dagger OE|\chi\rangle, \tag{25}$$

which is independent of all six angles. □

This result characterizes 72 stored angles from at most 66 output relevant coordinates, which does not assert exact rank 66. It also identifies six terminal rotations that can be omitted when only these observables are required, while the evaluated model retains its original parameterization.

5.5 Circuit resources and measurement implementation

Our quantum circuit has $qL = 24$ angle-encoding rotational gates, $3qL = 72$ trainable rotational gates, $qL = 24$ CNOTs, and a 64-amplitude statevector representation. Here, $L = 4$ repeated quantum circuit blocks. Through statevector operation, classical simulation supports direct reverse-mode differentiation.

While computing expectation values, every setting estimates all observables in a single basis. At most $2(72) = 144$ shifted settings are used for direct gatewise parameter-shift for the stored-angle feature Jacobian and $2qL = 48$ additional settings for derivatives of the re-uploaded inputs. To compute the expectation value, each setting needs to be run for a repeated number of shots.

Each bit donates an independent, separate sign factor, due to which an independent symmetric readout bit flip with probability p attenuates a weight-w product of Z variables by $(1-2p)^w$. Single-Z and pair-ZZ features therefore have factors $1-2p$ and $(1-2p)^2$, respectively. Together with Proposition 2, this specifies the measurement conventions for a future shot-based evaluation.

6 Training objective and experimental protocol

6.1 Phase-aware reconstruction and loss

From decoder outputs p_0, p_1, p_2, p_3, define

$$M_{fk} = 2\,\mathrm{sigmoid}(p_{0,fk}), \quad \Delta\phi_{fk} = 0.75\tanh(p_{1,fk}), \quad r_{fk}^{(R)} = 0.1\tanh(p_{2,fk}), \quad r_{fk}^{(I)} = 0.1\tanh(p_{3,fk}). \tag{26}$$

Using the stabilized noisy magnitude defined above, the reconstruction head gives

$$\widehat{S}_{fk} = m_{fk}\left[M_{fk}e^{i(\arg X_{fk}+\Delta\phi_{fk})} + r_{fk}^{(R)} + ir_{fk}^{(I)}\right]. \tag{27}$$

Bounded phase adjustment, magnitude scale Cartesian refinement, and magnitude modulation are merged using this reconstruction. During preprocessing, target and mixture joint amplitudes are rescaled.

For each sample, the composite objective is

$$\mathcal{L} = 0.30\mathcal{L}_{RI} + 0.20\mathcal{L}_{mag} + 0.05\mathcal{L}_{phase} + 0.20\mathcal{L}_{MR} + 0.25\mathcal{L}_{SI}. \tag{28}$$

The Cartesian and magnitude terms use mean Smooth-L_1 loss (transition parameter 1) on the real/imaginary components and stabilized magnitudes, respectively. The phase term is

$$\mathcal{L}_{phase} = \frac{1}{TF}\sum_{f,k} w_{fk}\left[1 - \cos(\arg\widehat{S}_{fk} - \arg S_{fk})\right], \tag{29}$$

where the clean-magnitude weight is normalized by its segment mean and clipped to $[0, 2]$. For STFT resolutions $(N_r, H_r) \in \{(256, 64), (512, 128), (1024, 256)\}$, define $A_r = \max(|\widehat{S}_r|, 10^{-7})$ and $B_r = \max(|S_r|, 10^{-7})$ elementwise. Then

$$\mathcal{L}_{MR} = \frac{1}{3}\sum_r \left[\frac{\|A_r - B_r\|_F}{\max(\|B_r\|_F, 10^{-7})} + \frac{\|\log A_r - \log B_r\|_1}{T_r F_r}\right]. \tag{30}$$

Each resolution uses a Hann window and centered frames. For mean-centered waveforms, the implemented stabilized training measure corresponding to SI-SDR [11] is

$$s_{\text{target}} = \frac{\langle \widehat{s}, s \rangle}{\|s\|_2^2 + \epsilon} s, \qquad \text{SISDR}_\epsilon = 10 \log_{10} \frac{\|s_{\text{target}}\|_2^2 + \epsilon}{\|\widehat{s} - s_{\text{target}}\|_2^2 + \epsilon}, \qquad \mathcal{L}_{SI} = -\text{SISDR}_\epsilon, \tag{31}$$

with $\epsilon = 10^{-8}$. Stabilization makes this a numerical extension of the ideal nonsilent definition. Low-SNR weights are 1.50, 1.30 and 1.15 at -5, 0 and 5 dB, and 1.00 at higher SNRs. For a batch, every component is reduced as $\sum_b w_b \ell_b / \max(\sum_b w_b, 10^{-8})$, rather than an unnormalized weighted sum. The training sampler uses inverse SNR-frequency sampling with replacement. These are distinct sampling and loss-weighting operations.

6.2 Data and distribution shifts

The mixture protocol uses 16-kHz VCTK-derived clean speech and environmental recordings from ESC-50 [12, 13]. The original manifests contain 15,000 training, 2,000 validation and 2,000 test records. Removing siren and helicopter from training and validation gives 12,175 and 1,644 records; selecting these categories in test gives 276 records. Training uses eight-second crops or padding. The unit of acoustic evaluation is a mixture; training seeds provide the repeated model fits.

The unseen-noise evaluation combines the two held-out categories with 23 speakers represented in the training/validation manifests. The double-OOD evaluation instead uses seven identities absent from those manifests: p276, p277, p278, p279, p282, p286 and p287. It contains 276 distinct clean source paths and eight distinct noise recordings. The same eight noise filenames occur in the unseen-noise test, so the two evaluations test different speech conditions against a common noise pool. Both span $\{-5, 0, 5, 10, 15, 20\}$ dB SNR.

The source-manifest audit found no cross-split overlap in sample identifiers, clean-source paths or noise-source paths. Speaker overlap is retained in the unseen-noise condition and removed at the manifest level in the double-OOD condition. The experiments use custom remixes with the inclusion rules above.

6.3 Training stages and comparison scope

For each seed, we pretrain the unconditioned backbone for at most 20 epochs, then warm up each conditioner with the frozen backbone for at most 8 epochs, followed by fine-tuning the conditioned system for at most 35 epochs. The start of each arm is from the seed's shared pretrained backbone. In the training configurations, the batch size is 8, weight decay of 10^{-4}, and gradient clipping at 5.0 with enabled mixed precision.

Our referenced implementation uses the AdamW optimizer with ReduceLROnPlateau, a factor of 0.5 with a patience of two validation epochs. In the case of paired arms, the conditioner shell resets identically:

- Xavier-uniform input weights
- Zero input biases
- Unit/zero layer-normalization affine parameters
- Output weights with standard deviation 10^{-3}
- Zero output biases and $\kappa = -2.5$

Projection learning rate and backbone learning rates are 2×10^{-4} and 10^{-4}, respectively. The classical and quantum core rates are 8×10^{-4} and 6.5×10^{-4}, respectively. The comparison uses a shared staged protocol with these separately chosen core learning rates. The unconditioned checkpoint is labeled a *pretrained reference* because it lacks the later backbone fine-tuning received by the conditioned systems. The primary comparison is between the two conditioned models; the pretrained model provides context for the training stages.

6.4 Seed inclusion and numerical analysis

All empirical tables use $\mathcal{R} = \{17, 23, 123\}$. This set was selected retrospectively after outcomes were available; an additional less favorable archived run is omitted without a documented outcome-independent invalidation criterion. The resulting comparisons are descriptive of the selected runs, rather than a prespecified all-run confirmation.

The reported metrics are SI-SDR, wideband PESQ [14] and ESTOI [15]. For metric m, seed r and mixture i, the paired effect is

$$d_{ri}^{(m)} = m(\widehat{s}_{ri}^{Q}, s_i) - m(\widehat{s}_{ri}^{C}, s_i), \quad \overline{d}_r^{(m)} = \frac{1}{276}\sum_i d_{ri}^{(m)}, \quad \overline{d}^{(m)} = \frac{1}{3}\sum_{r\in\mathcal{R}} \overline{d}_r^{(m)}. \tag{32}$$

For both absolute scores and paired effects, the displayed dispersion is the sample standard deviation of three seed means,

$$s_d^{(m)} = \sqrt{\frac{1}{3-1}\sum_{r\in\mathcal{R}} (\overline{d}_r^{(m)} - \overline{d}^{(m)})^2}. \tag{33}$$

This dispersion describes variation across three model fits. The analysis uses matched mixture-level differences within each seed and reports seed-level summaries; no confirmatory significance claim is made.

7 Results

7.1 Overview of enhancement results

For both distribution-shift conditions, quantum conditioning achieves the highest ESTOI, SI-SDR, and PESQ among all evaluated models (Table 1). Our primary comparison with the classical approach achieves a positive mean difference for every seed and metric (Table 2). Before aggregation, pairing, acoustic labels, and numerical summaries were checked for each seed model.

Table 1: Absolute enhancement scores, mean ± sample standard deviation of the three selected seed means. Each seed evaluates 276 mixtures. The pretrained reference has a shorter training budget.

Evaluation	Model	SI-SDR (dB)	PESQ	ESTOI
Unseen noise	Pretrained reference	15.18303 ± 0.24309	2.03437 ± 0.01155	0.73680 ± 0.00238
Unseen noise	Classical conditioner	16.08681 ± 0.16719	2.21134 ± 0.02890	0.75571 ± 0.00096
Unseen noise	Quantum conditioner	16.13653 ± 0.17241	2.22373 ± 0.03230	0.75660 ± 0.00130
Double OOD	Pretrained reference	15.36493 ± 0.22732	2.06558 ± 0.01723	0.75310 ± 0.00247
Double OOD	Classical conditioner	16.34827 ± 0.13665	2.24856 ± 0.02632	0.77303 ± 0.00020
Double OOD	Quantum conditioner	16.40531 ± 0.11671	2.26112 ± 0.02759	0.77362 ± 0.00060

Table 2: Paired quantum-minus-classical differences for each selected training seed and their mean ± sample standard deviation. Dispersion describes three seed averages; it is not a confidence interval.

Evaluation	Seed	Δ SI-SDR (dB)	Δ PESQ	Δ ESTOI
Unseen noise	17	+0.102994	+0.016795	+0.001321
Unseen noise	23	+0.036113	+0.013190	+0.000743
Unseen noise	123	+0.010047	+0.007201	+0.000591
Unseen noise	Mean ± SD	0.049718 ± 0.047944	0.012395 ± 0.004846	0.000885 ± 0.000385
Double OOD	17	+0.105636	+0.017862	+0.001062
Double OOD	23	+0.033493	+0.011945	+0.000491
Double OOD	123	+0.031974	+0.007861	+0.000211
Double OOD	Mean ± SD	0.057034 ± 0.042097	0.012556 ± 0.005029	0.000588 ± 0.000434

7.2 Unseen-noise enhancement

The quantum-conditioned model reaches 16.13653 dB SI-SDR, 2.22373 PESQ and 0.756597 ESTOI in the selected-seed summary. Relative to the matched classical core, the mean differences are +0.049718 dB, +0.012395 and +0.000885. Each selected seed has a positive average difference for all three metrics. PESQ improves by approximately 0.0072–0.0168 across these seeds. The seed-wise presentation shows that the direction of the mean difference is shared across these runs, with run-dependent magnitudes.

The improvements over the classical conditioner occur at the same total stored parameter count and with the same twelve-feature interface. They therefore characterize a change in the nonlinear conditioning

map within the specified training protocol. The shorter-trained pretrained reference provides additional context for absolute enhancement quality.

7.3 Joint speaker and noise shift

On the seven-speaker double-OOD test, quantum conditioning reaches 16.40531 dB SI-SDR, 2.26112 PESQ and 0.773617 ESTOI. Its corresponding mean differences are +0.057034 dB, +0.012556 and +0.000588. All selected seeds again have positive average differences on each metric. The positive average differences extend to speech from seven identities absent from the training and validation manifests. Figure 2 summarizes the paired effects across both evaluation conditions.

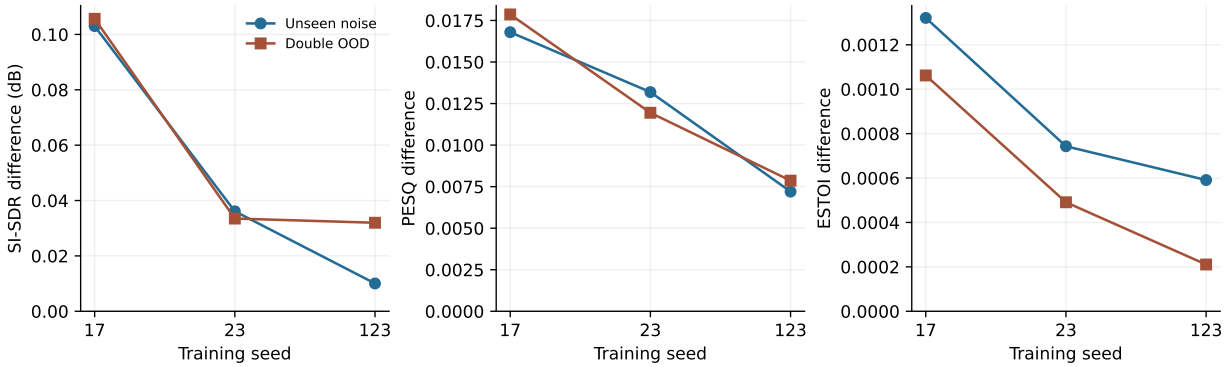


Figure 2: QNN-minus-classical mean differences for each selected seed. Each point averages 276 paired mixtures. These are seed means, not confidence intervals; all numerical results use seeds 17, 23 and 123. The selection scope in Section 6.4 applies to both evaluations.

These results describe mean enhancement performance for the reported seed set. The mean gains are small in absolute terms and are not uniform over mixtures or acoustic subgroups; listening benefits have not been assessed.

8 Discussion and future work

8.1 Global quantum conditioning for speech enhancement

The results support segment-level conditioning as a workable interface between a compact quantum model and a classical speech-enhancement backbone. The circuit processes six coordinates while its outputs modulate 128 bottleneck channels over the entire time–frequency grid. Consequently, its qubit count and number of ansatz gates remain fixed as segment length changes, although the classical backbone and feature pooling still depend on input size.

The quantum map combines periodic angle dependence with single-qubit and pairwise moments. Re-uploading gives a bounded higher-harmonic structure in the encoded coordinates, while pair moments permit non-factorized features at the output interface. The classical core supplies a useful stored-count comparison with a different feature structure. The observed differences concern these complete conditioning maps under their stated optimization settings; parameter-count matching does not equalize function classes or establish an entanglement-specific mechanism.

The measurement analysis complements the enhancement results by specifying how all twelve features can be collected in one basis and how their errors affect modulation. Equation (22) links the necessary shot budget to segment-level feature magnitude, learned modulation strength and projection norms. This gives a concrete starting point for designing measured implementations of the same interface.

8.2 Scope and next experiments

The present empirical contribution is an ideal-simulation comparison for the retrospectively selected runs described in Section 6.4. The two evaluations share a noise pool and cover two held-out noise categories. Their positive mean effects motivate a prospective evaluation with fixed seed inclusion, additional noise

recordings and a standard speaker-disjoint benchmark. The current evidence supports improvements within the reported comparisons, without establishing a hardware speedup or a general quantum advantage.

Future work will investigate correlated-feature and periodic classical conditioners, an equally fine-tuned unconditioned reference, and ablations of circuit depth, re-uploading and entangling gates. These experiments can clarify which representational choices contribute to enhancement. Shared-bitstring finite-shot evaluation and explicitly specified physical noise channels will test measured implementations, while listening studies can assess the perceptual importance of the metric changes. Such extensions build on the current architecture and mathematical analysis; their outcomes remain to be determined.

9 Conclusion

We presented a six-qubit variational bottleneck conditioner for phase-aware speech enhancement. The architecture converts multiscale global statistics into twelve quantum measurement features and applies them through bounded channel modulation. A classical conditioner with the same stored core parameter count provides a common-interface comparison. The accompanying analysis characterizes encoded-angle Fourier support, joint measurement statistics, gatewise derivatives and sensitivity of the bottleneck to moment-estimation error.

Across the reported seeds 17, 23 and 123, quantum conditioning improves mean SI-SDR, PESQ and ESTOI over the classical core in both unseen-noise and joint speaker–noise evaluations. These positive selected-run findings establish a concrete application of global quantum conditioning and provide a basis for broader controlled studies of quantum feature maps in speech enhancement.

A Numerical verification and reproducibility

An independent double-precision simulator checks the ordered circuit in Eq. (7). Across five random circuit instances, 360 stored-angle derivatives and 30 shared-input derivatives agree with central finite differences within 8×10^{-11}. Thirty encoded-coordinate Fourier slices have out-of-support amplitudes below 10^{-15}, and terminal redundant-angle derivatives are below 10^{-15}. All 250 sampled feature perturbations satisfy Proposition 3. These tests support implementation consistency; the propositions are established by their proofs.

The complex64 reference implementation agrees with the independent simulator within 1.3×10^{-7} for statevectors and 1.9×10^{-7} for expectation values. Autograd derivatives agree with independently calculated gatewise derivatives within 8×10^{-7}. Instantiated models confirm the parameter counts and pass CPU forward/backward checks on a 4,097-sample waveform, including finite outputs and nonzero conditioner-core gradients.

The accompanying verification package contains the per-mixture records used in the tables, aggregation scripts, circuit checks and the model source used for definition-level verification. Recomputed mean differences agree with the supplied summary values within 10^{-12}. The selected checkpoints and exact executed remix/evaluation runners have not been independently reproduced; definition-level checks and verification of metric arithmetic should be distinguished from a complete reproduction of training and inference.

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